FRAMEWORK REFERENCE

Agentic Systems Design

The AWD → SDD → APA Pipeline

A formalized framework for translating human task execution into agent-executable, spec-governed, and fully automated workflows.

Overview

Agentic Systems Design (ASD) is a three-layer framework that defines how human work is progressively translated, specified, and automated using AI agents. Each layer builds on the one before it — you cannot automate what is not specified, and you cannot specify what is not mapped.

AWD
Agentic Workflow Design
Map the work

→

SDD
Spec-Driven Development
Contract the work

→

APA
Agentic Process Automation
Automate the work

The three layers share a single progression: work moves from being human-executed, to being agent-readable, to being agent-governed, to being fully unattended — with humans present only at the edges.

Layer 1 — Agentic Workflow Design (AWD)

AWD
Agentic Workflow Design

The structured decomposition of a human task into a sequence of discrete, agent-executable steps — with defined triggers, transitions, branching logic, and handoffs.

Core Act: Translate human tasks into an agent-readable flow
Artifact: Flow map / task graph
Human Role: Designer — the human architects the map
Failure Mode: Gaps in task coverage (steps assumed, not defined)
Maturity Signal: All steps in the workflow are named and sequenced

"The workflow exists independent of who executes it — human or machine."

Key Principle

AWD is not automation yet. It is a formalized map of how work flows — one that a human could still walk through manually, but is now expressed in a form that agents can understand and traverse. The workflow precedes the executor.

What AWD Produces

- A trigger: what event or condition starts the workflow
- A step sequence: discrete, named actions in order
- Branching logic: decision points and their conditions
- Handoff definitions: where control passes between agents or humans
- Terminal states: what "done" looks like at the workflow level

Layer 2 — Spec-Driven Development (SDD)

SDD
Spec-Driven Development

The practice of expressing each workflow step as a formal behavioral contract that defines inputs, outputs, constraints, and acceptance criteria — making agent execution verifiable.

Core Act: Contract each step for correctness and verifiability

Artifact: Spec / contract document

Human Role: Specifier and approver — the human defines what "correct" means

Failure Mode: Ambiguous contracts (steps defined but not bounded)

Maturity Signal: All steps have bounded inputs, outputs, and acceptance criteria

"A spec is not documentation of what was built. It is the authority against which what gets built is judged."

Key Principle

The spec is the bridge between human intent and agent execution. It is not the code, and it is not the prompt — it is the contract that makes both verifiable. SDD is discipline-neutral: it applies to any domain where correctness must be defined before execution begins.

What a Spec Contains

- Inputs — what data or artifacts the step consumes
- Outputs — what the step must produce
- Constraints — rules that must hold during execution
- Acceptance criteria — the conditions under which the output is approved
- Rejection conditions — what constitutes a failed execution of this step

Layer 3 — Agentic Process Automation (APA)

APA
Agentic Process Automation

The execution state in which an agent fulfills a spec-governed workflow end-to-end, with humans present only at input definition and output validation.

Core Act: Execute the contracted workflow, human at edges only
Artifact: Running agent pipeline
Human Role: Input provider and output validator — human at edges only
Failure Mode: Unchecked agent drift (execution diverges from spec)
Maturity Signal: All steps are executed without human intervention

"The human is not removed from the system. They are moved to its edges."

Key Principle

APA is distinct from scripted automation. The agent handles ambiguity, makes decisions within defined constraints, and self-corrects — but only within the boundaries the spec allows. The spec is the leash; the workflow is the track; the agent is the runner.

What APA Requires

- A complete AWD map — no unnamed or assumed steps
- A complete SDD contract — every step is bounded
- An agent capable of self-correction within spec boundaries
- An input surface — how the human initiates the pipeline
- An output validation surface — how the human approves the result

Comparison at a Glance

Dimension	AWD	SDD	APA
Primary Question	What needs to happen?	What counts as correct?	Who executes it?
Artifact Produced	Flow map / task graph	Spec / contract document	Running agent pipeline
Human Role	Designer	Specifier / approver	Input provider / output validator
Failure Mode	Gaps in task coverage	Ambiguous contracts	Unchecked agent drift
Maturity Signal	All steps are named	All steps are bounded	All steps are unattended

Adoption Model

Teams rarely skip a layer cleanly. The natural progression is:

- Adopt AWD first — make the work visible and traversable
- Adopt SDD second — make the work verifiable and bounded
- Adopt APA third — make the work unattended and repeatable

Attempting APA without SDD produces agent drift. Attempting SDD without AWD produces contracts for undefined steps. The layers are load-bearing — each one supports the next.

Teaching Analogy — Building a House

The AWD is the construction schedule: what gets done, by whom, in what order. The SDD is the blueprint and building code: the contract each step must satisfy. The APA is when the crew shows up, reads the blueprint, and builds the house while you are away — and you only check when the keys are handed over.

Umbrella Term

Agentic Systems Design

ASD

The complete discipline covering workflow mapping, spec contracting, and agent-executed automation.

When referencing the full framework, use Agentic Systems Design (ASD) or the AWD → SDD → APA pipeline. When teaching individual layers, use the formal abbreviations — AWD, SDD, and APA — to reinforce the layered dependency.